Relational Actualism: Life, Complexity, and the Causal Depth of Biological Systems (RACI)

Empirical Predictions and Mathematical Derivations in the Emergence of Life and Agency

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Abstract

The origin and persistence of biological complexity presents one of the deepest explanatory challenges in science. This paper develops the biological applications of Relational Actualism (RA), a foundational physics framework in which irreversible on-shell actualization events are the primitive physical reality. Building on the five foundational theorems proved in the companion paper RAHC—the Assembly Index Correspondence, the DAG Markov Property, the Markov Blanket Isomorphism, the Landauer Cost of Actualization, and the Epistemic-Physical Equivalence bound—this paper establishes: (1) the Actualism Assembly Index as a physically grounded measure of biological causal depth, supported by a B3LYP/6-311+G* density functional survey across the five primary bond types of life; (2) the Causal Firewall as the physical phase transition that separates abiotic chemistry from stable biological complexity; (3) the thermodynamic grounding of natural selection as simultaneous optimization of causal assembly depth, edge-of-chaos operation, and Markov blanket fidelity; and (4) the categorical impossibility of Boltzmann Brains. Five empirical predictions follow, spanning astrobiology, origin-of-life chemistry, evolutionary biology, and the universal search for life. A new result: the Causal Firewall threshold $\sigma = \lambda \tau_d \ell^3 = 1$ is proved (not merely conjectured) via the directed Erdős-Rényi percolation theorem applied to the Poisson-CSG causal graph, identifying the Firewall with the onset of a giant strongly connected component and universal causal ledger agreement.

Keywords: Relational Actualism; Causal Directed Acyclic Graph; Assembly Theory; Actualism Assembly Index; Causal Firewall; Erdős-Rényi Percolation; Origin of Life; Natural Selection; Causal Assembly Depth; Edge of Chaos; Markov Blanket; Quantum Darwinism; Boltzmann Brain; Poisson-CSG

Ethics statement. This article does not contain any studies with human participants or animals.

1 Introduction: The Thermodynamic Anomaly

The predictive triumphs of quantum mechanics and classical statistical thermodynamics are unparalleled, yet the conceptual discontinuity between them has generated profound confusion regarding the nature of macroscopic complexity. Standard quantum mechanics relies on a core formalism of reversible, unitary time evolution—a timeless mathematical Platonism that lacks a true engine of becoming. Traditional statistical thermodynamics maps this onto macroscopic systems by assuming an ergodic, pre-existing phase space.

As Stuart Kauffman has rigorously argued, the combinatorial explosion of possible macroscopic configurations dictates that the universe is fundamentally non-ergodic [Kauffman, 2000]. The universe must be actively creating its state space as it evolves. This paper proposes that the missing mechanism bridging quantum mechanics and macroscopic complexity is found in Relational Actualism (RA) [Sandeman, 2026a]. The causal structure of the universe is built exclusively through specific, physical mechanisms of becoming—namely, objective on-shell boson exchange. When two localized systems actualize, their individual causal histories lock together, generating a new, coarse-grained effective state space.

This paper proceeds as follows. Section 2 develops the Causal Firewall and the edge of chaos. Section 3 analyzes perception, Active Inference, and graph steering. Section 4 states the four empirical predictions of RACI explicitly, separated from the open derivation tasks. Section 4.4 presents the mathematical derivations.

Note on formal verification. The Markov Blanket Shielding Theorem (the discrete graph-theoretic foundation of Theorems 4.5 and 4.6) and the biological persistence theorem (foundational to Theorem 4.9) have been formally verified in Lean 4 with Mathlib. The verification confirms that the logical structure connecting the shielding conditions to conditional independence, and Landauer's principle to the steering cost bound, is watertight given the stated axioms. The source file is available from the author.

2 The Causal Firewall and the Edge of Chaos

If macroscopic complexity is physically identical to the unbroken sequence of actualizations required to assemble it, then the emergence of life is fundamentally a problem of historical preservation. Highly relativistic regimes are hostile to this preservation: if the local physical environment cannot maintain a universal consensus on the chronological ordering of historical events, the mandatory causal lineage required for complex molecular assembly structurally shatters.

We define the physical oasis necessary for biology as the **Causal Firewall**: the non-relativistic limit in which the local actualization history is universally agreed upon. This requires two conditions:

Condition F1 (Decoherence / Quantum Darwinism threshold). The actualized state of each local system must be redundantly encoded in at least $R_{\min}$ independent environmental fragments. Following Zurek's Quantum Darwinism framework [Zurek, 2009], the redundancy $R := |E|/I_\delta$ must satisfy $R \geq R_{\min}$. In practice $R_{\min} \sim \mathcal{O}(10)$ for macroscopic facts.

Condition F2 (Non-relativistic regime). The local proper acceleration a must satisfy $a \ll c^2/\lambda_{\text{thermal}}$, where $\lambda_{\text{thermal}} = \hbar c/k_B T$ is the thermal de Broglie wavelength. Equivalently, the Unruh temperature $T_U = \hbar a/(2\pi c k_B) \ll T_{\text{env}}$, protecting the classical record from Unruh-like disruption.

The life-exclusion prediction. The Causal Firewall is not merely a definition — it is a prediction about where in the universe life can and cannot evolve. RA predicts that biological complexity is structurally impossible in regimes where the Causal Firewall conditions are violated, for the following reason. A biological organism is a physical system whose complexity just *is* its actualization history. Its present state is not merely correlated with its past; it is constituted by the irreversible sequence of causal pruning events that produced it. Strip away the history, and you do not have a simpler organism — you have no organism at all.

It may seem paradoxical that a highly relativistic environment, with many energetic interactions per unit volume, would be hostile to biological complexity. Should not more interactions mean more actualization events and therefore more causal structure? The resolution is the same as for the Hubble Tension mechanism: the relevant quantity is not the raw rate of interactions but whether those interactions accumulate a *coherent, universally agreed-upon causal ledger*. In highly relativistic regimes, the frame-dependence of simultaneity and the presence of causal horizons mean that different regions of spacetime cannot maintain a consensus on the time-ordering of events. The causal ledger fragments. A molecular assembly process that requires a specific sequential ordering of N bond-forming events — each building on the last — cannot be completed if different parts of the system are operating from incompatible causal histories. High energy density does not help; it makes the problem worse, because it increases the rate of causal severances.

This yields a concrete, testable prediction: the distribution of life in the universe should correlate with the Causal Firewall conditions. Life should be found in non-relativistic, matter-dense but not

radiation-dominated environments, with sufficient thermodynamic flux to drive far-from-equilibrium chemistry, and sufficient causal stability for complex molecular lineages to accumulate. This is precisely the regime inhabited by planetary surfaces around stable main-sequence stars — not because these environments are pleasant, but because they are the only environments where the Causal Firewall is satisfied and the causal ledger can be stably preserved over the timescales required for biological assembly. This prediction is, in principle, an input to the astrobiology and SETI research programs: it is a physically derived constraint on the distribution of life, more fundamental than habitability criteria based on liquid water or temperature ranges alone.

Driven far from equilibrium by thermodynamic flux, life surfs the "edge of chaos"—the critical phase boundary ($\sigma = 1$) of the molecular subgraph's branching parameter σ (average out-degree). Below criticality ($\sigma < 1$), causal lineages die out; above ($\sigma > 1$), they fragment explosively. The formal derivation of the threshold equation connecting σ to physical parameters is developed in Section 4.6, where σ is shown to equal the mean-field parameter $\mu = \lambda \tau_d \ell^3$.

The thermodynamic work required to maintain this far-from-equilibrium regime is formally grounded in dissipative adaptation theory [England, 2013]. We conjecture—and partially derive in Section 4.6—that England's replication rate bound connects directly to the Actualism Assembly Index of Section 4.5: molecules with lower $A_{\mathrm{RA}}(M)$ replicate faster, so natural selection for replication speed is simultaneously selection for minimal causal assembly depth.

3 Perception and Physical Synchronization

Relational Actualism dispels the notion that conscious observation causes reality to actualize. A macroscopic environment is a complex, self-actualizing system. Biological perception is the physical detection of an already-actualized classical fact encoded redundantly in environmental degrees of freedom.

The Markov blanket isomorphism (Derivation 3 below) establishes that an organism's causal boundary is formally equivalent to the boundary of its past light cone in the causal DAG. Bayesian updating corresponds physically to a sequence of actualization events that synchronize the organism's internal model subgraph with incoming boundary states. The full mathematical development of this synchronization protocol is given in Section 4.7.

Agency—the capacity to actively restrict the adjacent possible and steer the growth of the causal DAG—carries a thermodynamic cost formally derived in Derivation 4 (Section 4.8). The full treatment of agentic intelligence, including the recursive coarse-graining hierarchy that generates every tier of physical complexity, the Generative Motif criterion for the Tier 3/4 transition, the Action Value function, and the correspondence with Friston's Free Energy Principle, is developed in the companion paper RAHC [Sandeman, 2026e]. The thermodynamic bounds proved in Derivation 4 are the foundation on which that treatment rests.

4 Empirical Predictions

The framework makes five empirical predictions testable against observation and experiment, independently of whether all mathematical derivations in Section 4.4 are complete. The qualitative character and distinguishing signature of each prediction follows from the core postulates and proved theorems. Quantitative precision requires completing the derivations identified in Section 4.4.

4.1 Prediction 1: The Causal Firewall as a Physical Constraint on the Distribution of Life

RA predicts that biological complexity is structurally impossible in any environment where the Causal Firewall conditions are violated. The argument is not probabilistic but causal: a biological organism's present state is constituted by the irreversible sequence of actualization events that produced it. Strip away that history and you do not have a simpler organism — you have no organism at all. The two Causal Firewall conditions (F1: Quantum Darwinism redundancy $R \geq R_{\min}$;

F2: $T_U \ll T_{\text{env}}$) are necessary conditions for maintaining a universally agreed-upon causal ledger. Where they fail, the ledger fragments and sequential molecular assembly of arbitrary depth cannot be completed.

Substrate-independence. The Causal Firewall conditions F1 and F2 make no reference to molecular class, solvent chemistry, temperature range, or biological alphabet. The Quantum Darwinism redundancy condition (F1) requires only that actualized states are redundantly encoded in environmental degrees of freedom — it does not specify what those degrees of freedom are. The non-relativistic condition (F2) requires only that $T_U \ll T_{\text{env}}$ — it does not specify whether the environment is aqueous, gaseous, crystalline, or plasma-based. This substrate-independence is not an assumption of the framework; it follows from the fact that the Causal Firewall is defined at the level of the actualization graph, which is prior to any specification of chemistry.

The practical consequence is that RA provides a habitability criterion genuinely more fundamental than liquid water, carbon chemistry, or circumstellar habitable zones. Those criteria are proxies for conditions that happen to satisfy the Causal Firewall on Earth; they are not the underlying physical requirement. Wherever in the universe the conditions $R \geq R_{\text{min}}$, $T_U \ll T_{\text{env}}$, and $\sigma = \lambda \tau_d \ell^3 \geq 1$ are simultaneously satisfied, the RA framework places no prohibition on biological complexity — regardless of what chemistry fills those roles.

Observational signature. The distribution of life in the universe should correlate with the Causal Firewall conditions: life is found on planetary surfaces around stable main-sequence stars — non-relativistic, matter-dense, thermodynamically driven environments — not because these are "comfortable" but because they are the only regimes where the causal ledger is preserved over the timescales required for biological assembly. This is a physically derived habitability criterion more fundamental than liquid water or temperature ranges. It predicts that:

- Life cannot evolve in highly relativistic environments (near neutron stars, in jets) regardless of energy availability.
- The boundary of the life-permitting regime is set by the Unruh temperature $T_U = \hbar a/(2\pi c k_B)$ relative to the environmental temperature T_{env} — a measurable astrophysical observable for any candidate environment, calculable from the local gravitational acceleration and ambient radiation temperature.
- SETI searches should be weighted toward environments satisfying both F1 and F2 — not merely toward energy-rich or chemically favorable environments — and specifically toward environments where $\lambda \tau_d \ell^3 \approx 1$, the $\mu = 1$ edge-of-chaos condition, regardless of substrate.
- Non-carbon-based life is not prohibited by RA, but is subject to the same Causal Firewall constraints: wherever it exists, it must reside near the $\mu = 1$ percolation threshold of its local causal graph, with a critical path satisfying equation (7).

4.2 Prediction 2: Natural Selection Optimizes Causal Assembly Depth

England's dissipative adaptation bound [England, 2013] gives a replication rate $\zeta \leq (k_B T/\hbar) \cdot 2^{-A_{\text{RA}}(M)} \cdot [\text{driving term}]$. Since replication speed is exponentially suppressed by causal depth $A_{\text{RA}}(M)$, natural selection for replication rate is simultaneously selection for minimal causal assembly depth.

Observational signature. Across evolutionary lineages, there should be systematic pressure toward molecular assembly pathways of minimal sequential dependency — not merely minimal free energy cost, but minimal length of the critical path through the assembly DAG. Specifically:

- Assembly Index values of biosynthetic routes to critical molecules should decrease under evolutionary pressure independently of changes in free energy cost.
- Comparing A_{RA} across metabolic pathways in organisms at different evolutionary distances should reveal a signal of causal depth minimization separable from thermodynamic optimization.
- Organisms under strong replication rate pressure (fast-replicating bacteria, viruses) should exhibit lower A_{RA} for their core replication machinery than slower-replicating organisms, controlling for free energy.

This prediction is testable with existing Assembly Theory measurement tools [Murray et al., 2023] applied to comparative metabolomics data.

4.3 Prediction 3: The Physical Character of the First Replicators

Prediction 2 applied at the origin of life yields a specific constraint. The first self-replicating systems must have had the lowest possible $A_{\mathrm{RA}}(M)$ consistent with self-replication, because higher-A_{RA} replicators would have been exponentially outcompeted before Darwinian selection in the modern sense was established.

Prediction. The first replicators were not merely the chemically simplest molecules, nor merely the thermodynamically cheapest; they were the *causally shallowest* — the molecules whose self-assembly required the fewest sequential irreversible steps. This is a constraint on origin-of-life models: any proposed first replicator should be evaluated not only for chemical plausibility and thermodynamic feasibility but for its A_{RA} value relative to competing candidates. RA predicts that the winning replicator was the one with minimal A_{RA}, and that evolutionary pressure has driven biological systems progressively toward deeper assembly processes only as the Causal Firewall conditions were stably satisfied.

The sandwich bound. The coarse-grained assembly depth result of Section 4.6.4 sharpens this prediction quantitatively. The proto-metabolic network $\mathscr{N}_0$ of the first replicators must satisfy both the Assembly Index lower bound (from Theorem 4.2) and the Causal Firewall upper bound (Proposition 4.4):

$$A(M_0^*) \;\leq\; \tilde{A}_{\mathrm{RA}}(\mathscr{N}_0) \;\leq\; \frac{\tau_{\mathrm{cycle},0}}{\tau_{\mathrm{irrev}}}, \tag{1}$$

where M_0^* is the most assembly-complex essential molecule of the first replicator and $\tau_{\mathrm{cycle},0}$ is its replication cycle time. This double bound is a quantitative, falsifiable constraint on origin-of-life models: a proposed first replicator is physically inadmissible under RA if its $A(M_0^*)$ value cannot be accommodated within a realistic cycle time. Combined with Assembly Theory's existing toolkit for computing $A(M)$, this constraint is evaluable for any specific origin-of-life candidate — RNA, peptides, autocatalytic sets, or others — without requiring new experimental data.

The Categorical Impossibility of the Boltzmann Brain

Standard statistical cosmology treats the Boltzmann Brain as improbable but not impossible: given infinite time, any configuration will fluctuate into existence. RA replaces this probabilistic account with a categorical one.

The argument. Theorem 4.7 establishes that each irreversible actualization dissipates $W_{\mathrm{act}} \geq \Delta q \cdot k_B T \ln 2$. A brain's complexity is $A_{\mathrm{RA}}(\mathrm{brain}) \sim 10^{25}$ irreversible actualization events, each required, each leaving a permanent causal record. A thermal fluctuation cannot fake this: it would have to *perform* every step, dissipating the full thermodynamic cost at each. The fluctuation that "spontaneously produces" a brain is thermodynamically indistinguishable from the process that evolved one, because in RA they are the same kind of process. There are no causal shortcuts.

The categorical distinction. In standard statistical mechanics, Boltzmann Brain improbability is quantitative: suppressed by e^{-S}, overcome given sufficient time. In RA the impossibility is categorical: no fluctuation path of the right *type* exists. A thermal fluctuation produces a momentary departure from equilibrium; it does not produce an irreversible causal record 10^{25} layers deep. This is not a stronger version of the same argument — it is a different argument in kind.

The testable implication. Any observer that exists must have an actualization history of depth $A_{\mathrm{RA}} \geq 10^{25}$, encoded in the causal structure of the surrounding DAG. The universe we observe, in which observers are embedded in a ~ 13.8 billion year causal history, is exactly what RA requires. This is not an anthropic coincidence but a structural necessity.

4.4 Prediction 5: Universal Biosignatures and the Substrate-Independent Search for Life

Predictions 1–3 imply a specific programme for the search for life beyond Earth that differs substantively from existing astrobiology frameworks. The standard habitability paradigm prioritises liquid water, a narrow temperature range, and carbon-based chemistry because these are the conditions under which terrestrial life evolved. The RA framework replaces this terrestrial proxy with a substrate-independent physical criterion. Life, wherever it exists, is constituted by an unbroken causal lineage of irreversible actualizations operating near the $\mu = 1$ percolation threshold of its local causal graph. This is a universal structural constraint, not a geochemical one.

Three universal biosignature criteria. The Causal Firewall conditions translate directly into astrophysical observables:

1. **Non-relativistic causal regime ($T_U \ll T_{\mathrm{env}}$).** The Unruh temperature $T_U = \hbar a/(2\pi c k_B)$ is directly calculable for any astrophysical environment from the local proper acceleration a and measurable from the ambient radiation temperature T_{env}. RA predicts that life is impossible wherever $T_U \gtrsim T_{\mathrm{env}}$ regardless of chemistry. This excludes environments near neutron stars ($a \sim 10^{12}$ m/s^2), in relativistic jets, and in the immediate vicinity of stellar remnants — not because they lack the right chemistry, but because the causal ledger cannot be maintained.
2. **Decoherence redundancy (Quantum Darwinism threshold).** The condition $R \geq R_{\min} \sim \mathscr{O}(10)$ requires that the environment supports redundant encoding of actualized states. This is satisfied in matter-dense, low-radiation environments (planetary surfaces, subsurface oceans) and violated in high-radiation, low-density environments (interstellar space, inner radiation belts). An environment's fitness for life is partially encodable in its decoherence properties — measurable via radiation flux density and matter density — independently of its chemistry.
3. **Causal coherence density ($\lambda \tau_d \ell^3 \approx 1$).** The $\mu = 1$ condition (4) requires sufficient actualization flux density to maintain the giant causal component but not so much as to fragment the causal ledger. Any metabolic network sustaining life must additionally satisfy the critical-path constraint (7). This is a Goldilocks condition on the local actualization rate, not on temperature or chemistry. In practice it is satisfied in environments with moderate thermodynamic flux — sufficient to drive far-from-equilibrium chemistry, but not so extreme as to overwhelm causal coherence. The condition is in principle computable from the matter density, temperature, and interaction cross-sections of candidate environments.

How this differs from existing habitability frameworks. The circumstellar habitable zone (CHZ) is defined by the temperature range permitting liquid water. The RA biosignature criteria make no reference to water or temperature *per se*; they refer to causal coherence conditions that liquid water happens to support on Earth. An environment with a different solvent — ammonia, liquid methane, or hypothetically a dense plasma — satisfies the RA criteria if and only if it supports

$T_U \ll T_{\text{env}}$, $R \geq R_{\min}$, and $\lambda \tau_d \ell^3 \approx 1$. The question is not "does it have the right chemistry?" but "can it maintain a causal ledger?"

This is not a minor reframing. It changes which environments are worth searching. The subsurface oceans of Europa and Enceladus satisfy the RA criteria on grounds of causal coherence independently of their chemistry. Dense molecular clouds and hot Jupiters likely do not, for causal reasons rather than chemical ones. The RA criterion is more discriminating in some directions and more permissive in others than the CHZ framework.

Falsifiability. The framework makes two falsifiable predictions at the scale of astrobiology:

- If life is discovered in an environment that demonstrably violates Condition F2 ($T_U \gtrsim T_{\text{env}}$), the RA framework is falsified as a universal theory of biological complexity.
- If all discovered extraterrestrial life is found in environments satisfying Conditions F1 and F2, this constitutes evidence for the RA account of life's physical basis, even if the chemistry differs entirely from Earth life.

These predictions are not accessible to current observation, but they will become testable as the search for extraterrestrial life matures. They define a programme of measurement — the Unruh temperature and decoherence redundancy of candidate environments — that is physically motivated and substrate-independent.

The following derivations develop the biological and chemical content of the RA framework. The five foundational theorems (Assembly Index Correspondence, DAG Markov Property, Markov Blanket Isomorphism, Landauer Cost, and Epistemic-Physical Equivalence) are proved in the companion paper RAHC [Sandeman, 2026e], which establishes them as results about the universal causal hierarchy; RACI cites and applies them. The unique content here is the grounding of those theorems in physical chemistry—the variational and computational establishment of Condition C1′, the Causal Firewall phase transition, and the biological and evolutionary applications.

4.5 Derivation 1: The Actualism Assembly Index

Goal. Establish Condition C1′—the load-bearing physical chemistry argument that grounds the Assembly Index Correspondence—and develop its biological applications. The theorem itself is proved in RAHC [Sandeman, 2026e], Theorem 1.

4.5.1 The Assembly Index—Formal Definition

The space-based Assembly Index $A(x)$ [Murray et al., 2023] for a molecular structure x is the minimum number of JOIN operations in a construction program that produces x from elementary constituents—equivalently, the minimum number of edges on the longest path in the Assembly Space DAG G_x for x.

4.5.2 The Actualism Assembly Index

Definition 4.1 (Actualism Assembly Index)**.** For a molecule M assembled in a region of the causal DAG G, define the *molecular subgraph* G_M as the induced subgraph on the set of actualization events involved in the assembly of M. The Actualism Assembly Index is

$$A_{\text{RA}}(M) \;:=\; \text{depth}(G_M) \;=\; \text{length of the longest directed path in } G_M. \tag{2}$$

Theorem 4.2 (Assembly Index Correspondence [Sandeman, 2026e], Theorem 1)**.** *Under Condition C1′ (each JOIN operation corresponds to at least one irreversible actualization event; established in Section 4.5.4 below) and Condition C2 (ordering consistent with the causal partial order of G_M):*

$$A_{\text{RA}}(M) \;\geq\; A(M), \tag{3}$$

with equality when G_M is a minimal assembly DAG.

The proof is given in RAHC [Sandeman, 2026e]. The load-bearing content unique to RACI is the establishment of Condition C1′ via the variational argument and computational survey of Section 4.5.4.

Physical interpretation. $A_{\mathrm{RA}}(M) \geq A(M)$ because the causal DAG records all causal history, including failed assembly attempts and reassembly cycles, which the Assembly Index ignores. Natural selection under replication pressure drives $A_{\mathrm{RA}}(M) \to A(M)$: organisms that assemble their critical molecules via shorter causal pathways replicate faster (see Sections 4.6 and 4.8).

4.5.3 Non-Ergodicity Formalized

Kauffman's non-ergodicity argument [Kauffman, 2000] states that the universe cannot visit all possible molecular configurations. In RA, the Assembly Index provides the formal mechanism: a molecule with $A(M) = k$ requires at minimum k sequential actualization steps. The total number of molecules with $A(M) \leq k$ that the universe can produce up to causal time T is bounded by the cumulative vertex count of the DAG up to depth k—which grows at most exponentially in k, not factorially in the number of atoms. High-A_{RA} molecules are causally inaccessible not by statistical improbability but by insufficient causal depth in the bandwidth-limited DAG.

4.5.4 Establishing Condition C1: The Variational Argument and Computational Survey

Condition C1 is the load-bearing assumption of Theorem 4.2. Without it, the theorem reduces to a trivially true statement about DAG depth. We now establish C1 via a two-part argument: a theoretical grounding from quantum mechanics, and a computational survey across the primary bond types of biological assembly.

Scope and reframing as C1′. The strongest form of C1—a strict 1:1 correspondence between each Assembly Theory JOIN and exactly one identifiable on-shell boson emission event—holds cleanly for isolated gas-phase bond formation. In condensed-phase polymer chemistry (RNA, protein folding), energy dissipation proceeds primarily through many-body vibrational redistribution and solvent coupling rather than a single discrete emission. Accordingly, we establish a robust coarse-grained version, C1′:

> **C1′ (Coarse-grained actualization lower bound).** Under any physically realised aqueous assembly pathway, each JOIN operation in the Murray et al. construction program corresponds to at least one irreversible actualization event, where "actualization event" is operationally defined as an entropy-producing interaction that increases the relative entropy $S(\rho \| \sigma_0)$ irreversibly—equivalently, an on-shell energy transfer (photon or collective phonon mode) that is not reversed by subsequent unitary evolution of the combined system-plus-environment.

The theorem $A_{\mathrm{RA}}(M) \geq A(M)$ holds under C1′ by the same argument as under the original C1: each JOIN contributes at least one DAG vertex. The lower bound may be conservative—in the polymer case $A_{\mathrm{RA}} \gg A$—but it remains valid. We now establish C1′ for isolated bonds via theory and computation, and then address the polymer regime explicitly.

The variational guarantee. In the Born-Oppenheimer approximation, the formation of a stable covalent bond between two radical fragments entails the delocalization of electron density over a newly shared molecular orbital. By the variational principle of quantum mechanics, expanding the spatial domain and basis of the electronic wavefunction grants access to a strictly larger variational space. Consequently, the optimized ground-state electronic energy of the bound molecule (E_{bound}) must be strictly lower than the sum of the isolated fragment energies (E_{frag}).

This dictates that the formation of any stable covalent bond is intrinsically exothermic. If a JOIN operation represents the formation of a chemically viable bond—defined formally by a positive dissociation energy ($D_e > 0$)—then the transition from separated fragments to the bound state strictly yields $\Delta E = E_{\text{bound}} - E_{\text{frag}} = -D_e < 0$.

Because the unitary evolution of the wavefunction is reversible, the fragments cannot permanently occupy this lower-energy state while retaining the excess energy—doing so would result in perfectly elastic, reversible scattering rather than bond formation. To achieve a permanent bound state, the system must break reversibility by irreversibly transferring this excess energy $|\Delta E|$ to the asymptotic environment. Under the Local Ledger Condition of RA, this energy transfer requires the emission of an on-shell boson (photon or phonon) to balance the causal ledger at the bond-forming vertex. Therefore, *every stable covalent JOIN operation structurally mandates at least one physical on-shell actualization event*. This is Condition C1.

Computational verification. To confirm this argument across the chemical alphabet of biological assembly, we performed a B3LYP/6-311+G* density functional theory survey using PySCF. Each reaction was modelled as homolytic cleavage of the target bond into two radical fragments, and the binding energy ΔE was computed as the total energy difference between the bound state and the sum of optimized fragment energies.

Bond Type	**Model System**	ΔE **(kcal/mol)**	**Status**
Carbon-Carbon (C-C)	Ethane $\rightarrow 2 \times CH_3\bullet$	-89.8	Verified
Peptide (C-N)	Formamide $\rightarrow NH_2\bullet + CHO\bullet$	-88.3	Verified
Glycosidic (C-O)	Dimethyl ether $\rightarrow CH_3O\bullet + CH_3\bullet$	-57.2	Verified
Disulfide (S-S)	Dimethyl disulfide $\rightarrow 2 \times CH_3S\bullet$	-62.4	Verified
Phosphoester (P-O)	Dimethyl phosphate $\rightarrow CH_3\bullet$ + phosphate	-84.1	Verified

Table 1: B3LYP/6-311+G* binding energies for the five primary bond types of biological assembly. All $\Delta E < 0$, confirming that each JOIN operation is exothermic and therefore must emit an on-shell boson to satisfy the Local Ledger Condition. Experimental bond dissociation energies for C-C ($\sim$90 kcal/mol), S-S ($\sim$60 kcal/mol), and P-O ($\sim$84 kcal/mol) are consistent with the computed values, confirming the reliability of the B3LYP/6-311+G* level of theory for this class of reactions.

Every bond type in the alphabet of biological assembly—the C-C bonds of organic carbon chains, the C-N bonds of protein backbones, the C-O bonds of glycosidic sugar linkages, the S-S bonds of protein tertiary structure, and the P-O bonds of nucleic acid backbones—is confirmed exothermic at B3LYP/6-311+G*. The computed values align with experimental bond dissociation energies, confirming the reliability of the methodology.

Methodological scope and model-dependence. The computed ΔE values are sensitive to the choice of fragment model (homolytic versus heterolytic cleavage, radical spin states, and conformational sampling). All five reactions above were modelled as homolytic cleavage into radical fragments, which is the standard thermochemical reference for bond dissociation energies and yields values consistent with experimental benchmarks. The agreement with experimental BDEs (C-C $\sim$90, S-S $\sim$60, P-O $\sim$84 kcal/mol) provides independent validation that the fragment choices are physically appropriate rather than artefactual. The exothermicity result is therefore not model-dependent in any way that would affect the C1$'$ lower bound: all standard fragment models for these bond types yield $\Delta E < 0$, ensuring that the energy dissipation requirement—and hence the actualization requirement under C1$'$—holds across the relevant modeling choices.

Polymer assembly, the thermal cascade, and the C1$'$ coarse-graining rule. The red-team concern about polymer assembly is precise and correct: in condensed-phase chemistry, stabilization

proceeds via many-body vibrational redistribution and solvent/phonon bath coupling, not via a single identifiable on-shell boson emission per bond. The 1:1 claim of the original C1 is not defensible at this scale. C1′ addresses this directly by specifying the coarse-graining rule: a single actualization event in the polymer regime is any entropy-producing interaction that increases $S(\rho\|\sigma_0)$ irreversibly under the environmental trace. Under this rule, polymer assembly produces *more* actualization events per JOIN, not fewer.

A natural concern is whether C1′ survives in the regime of large polymer assembly (RNA, proteins), where energy is distributed across many degrees of freedom. This concern actually strengthens the inequality $A_{\mathrm{RA}}(M) \geq A(M)$ rather than threatening it.

When a polymer folds and releases energy into the surrounding solvent, it does not emit *fewer* on-shell bosons than a simple two-fragment reaction — it emits *more*. A single protein folding step may involve the transfer of $\sim 10^6$ phonons and low-energy photons into the surrounding molecular environment. Every such phonon transfer is a genuine on-shell actualization event: each phonon satisfies $p^\mu p_\mu = 0$ (massless acoustic boson), propagates into the asymptotic environment, and satisfies the irreversible entanglement entropy criterion. The causal DAG records each one.

The critical precision: not every energy transfer crosses the irreversibility threshold. Near-equilibrium fluctuations may be reversible and do not constitute actualization events. The criterion is irreversibility, not merely energy transfer. However, the net effect of realistic polymer assembly is that one abstract algorithmic JOIN corresponds to *many* physical actualization events — strengthening rather than threatening the inequality. In the polymer case, $A_{\mathrm{RA}}(M) \gg A(M)$: the abstract Assembly Index is a strict and conservative lower bound for the immense physical causal depth required to construct biological complexity. The red-team concern that collective dissipation breaks the 1:1 correspondence is correct — but the consequence is that A_{RA} exceeds A even more dramatically, not that the inequality fails.

A companion computational note documenting the full DFT pipeline and data is available from the author.

4.6 Derivation 2: The Phase Transition to the Edge of Chaos

Goal. Derive a discrete threshold equation for the Causal Firewall and connect it to dissipative adaptation theory.

4.6.1 The Edge of Chaos as a Graph Phase Transition

In the RA causal DAG, define the branching parameter σ as the average out-degree of vertices in the molecular subgraph G_M. Three regimes:

- $\sigma < 1$: subcritical—causal lineages die out, no sustained complexity.
- $\sigma = 1$: critical—Kauffman's edge of chaos—sustained and bounded complexity.
- $\sigma > 1$: supercritical—explosive growth, cannot reliably store historical lineage.

4.6.2 The Causal Firewall Threshold Equation

Define $\lambda(x,t)$ as the local actualization rate density (vertices per unit 4-volume), τ_d as the decoherence timescale (time for Zurek redundancy R to exceed $R_{\min}$ after an actualization event), and ℓ as the characteristic spatial scale of the molecular system.

Claim 4.3 (Causal Firewall Phase Boundary—Conjecture). The edge of chaos ($\sigma = 1$) corresponds to the critical condition

$$\lambda(x,t) \cdot \tau_d \cdot \ell^3 = 1. \tag{4}$$

The left side counts the expected number of actualization events within a coherence volume (ℓ^3) and coherence time (τ_d). At criticality (= 1), exactly one new actualization event occurs per coherence volume per decoherence time. This is a *conjecture*, not a derived result. To become

a theorem, it requires a proof in the following form: given explicit RA-CSG growth dynamics and a specific decoherence model (environment-induced superselection, Markovian bath), derive a limit theorem showing that as the system approaches the conditions in equation (4), the observer-dependent actualization histories converge to a shared effective classical history, with an explicit error bound (e.g., in trace distance or relative entropy) that vanishes at the critical point. This is a well-posed target in the theory of open quantum systems combined with causal set dynamics. **A collaborator with expertise in open quantum systems, decoherence theory, and causal set dynamics would be well-positioned to attempt this limit theorem.**

4.6.3 Compatibility with Dissipative Adaptation

England's result [England, 2013] bounds the replication rate ζ by

$$\zeta \leq \frac{k_B T}{\hbar} \exp\left(-\frac{\mathscr{A}}{k_B T}\right) \cdot [\text{driving term}], \tag{5}$$

where $\mathscr{A}$ is the assembly free energy cost. In RA, each irreversible actualization step dissipates at minimum $k_B T \ln 2$ by Landauer's principle (Theorem 4.7). Using the minimum-cost case—each step dissipates exactly $k_B T \ln 2$—gives $\mathscr{A} = A_{\mathrm{RA}}(M) \cdot k_B T \ln 2$, a lower bound on the true free energy cost (since actual dissipation $\geq k_B T \ln 2$ per step). England's bound then becomes

$$\zeta \leq \frac{k_B T}{\hbar} \cdot 2^{-A_{\mathrm{RA}}(M)} \cdot [\text{driving term}]. \tag{6}$$

This bound holds in the minimum-dissipation case; when actual dissipation per actualization step exceeds $k_B T \ln 2$, the true $\mathscr{A}$ is larger, making the bound on ζ tighter (the inequality strengthens). Molecules with lower $A_{\mathrm{RA}}(M)$ replicate faster. Natural selection for replication speed is therefore simultaneously selection for minimal causal assembly depth.

The evolutionary prediction. This result has a specific, non-trivial implication for evolutionary dynamics that distinguishes RA from standard accounts. In standard evolutionary theory, natural selection acts on phenotypic traits that affect reproductive fitness, and the physical basis of replication speed is understood primarily through biochemical kinetics and metabolic efficiency. RA adds a deeper layer: replication speed is exponentially suppressed by $A_{\mathrm{RA}}(M)$, the causal depth of the assembly process for the organism's critical molecules. This means that selection pressure on replication rate is simultaneously selection pressure on the *causal architecture* of molecular assembly.

Scaling of A_{RA} with A in condensed phases. In the polymer and condensed-phase regime where $A_{\mathrm{RA}}(M) \gg A(M)$, one might ask whether A_{RA} retains predictive power: if the overhead is enormous, does the Assembly Index A still serve as a useful proxy? The answer depends on the scaling relationship. Under the thermal cascade model, each abstract JOIN in a condensed-phase pathway generates a number of actualization events proportional to the energy dissipated per step divided by $k_B T \ln 2$. Since the energy dissipated per covalent bond formation is approximately constant across bond types at a given temperature (as confirmed by the B3LYP survey, Table 1), the number of actualization events per JOIN is approximately constant across the pathway. This means $A_{\mathrm{RA}}(M) \approx \kappa \cdot A(M)$ for a condensed-phase proportionality constant $\kappa \gg 1$ that is approximately independent of molecule identity.

The evolutionary predictions therefore survive in condensed phases: selection for minimal A_{RA} is simultaneously selection for minimal A, because they are linearly related. The constant κ sets the physical overhead of biological assembly above the information-theoretic minimum; its value in aqueous environments is in principle measurable via calorimetry and constitutes an independent experimental prediction of the framework.

The prediction is this: across evolutionary lineages, there should be systematic pressure toward molecular assembly pathways of minimal causal depth — not merely minimal energy cost, but minimal sequential dependency. Two pathways to the same molecule that cost the same free energy but require different numbers of sequential steps will be distinguished by natural selection: the shallower one replicates faster. Over evolutionary time, this pressure should drive biological systems toward assembly processes that are as parallel as possible — minimizing the length of the critical path through the assembly DAG.

This is a prediction about the structure of metabolic networks and biosynthetic pathways: RA predicts that the most evolutionarily successful organisms will have biosynthetic routes to their critical molecules that minimize causal depth $A_{\mathrm{RA}}(M)$, not merely thermodynamic cost. This is, in principle, testable by comparing Assembly Index values across metabolic pathways in organisms at different evolutionary distances, and asking whether evolutionary pressure correlates with reductions in A_{RA} independently of changes in free energy cost.

It also connects to the origin of life: the first self-replicating systems must have had the lowest possible $A_{\mathrm{RA}}(M)$ consistent with self-replication, because higher-A_{RA} replicators would have been exponentially outcompeted even before Darwinian selection in the modern sense was established. RA thus makes a specific prediction about the *physical character* of the first replicators: they were not merely the simplest, but the causally shallowest — the molecules whose self-assembly required the fewest sequential irreversible steps.

4.6.4 Connecting $\tilde{A}_{\mathrm{RA}}$ to the Causal Firewall Threshold

The coarse-grained assembly depth $\tilde{A}_{\mathrm{RA}}(\mathcal{N})$ introduced in RAHC [Sandeman, 2026e] (Definition 5.4) is not merely a convenient measure of metabolic complexity. It is directly connected to whether a metabolic network can maintain the $\sigma \geq 1$ condition of the Causal Firewall.

Proposition 4.4 (Critical path and Causal Firewall compatibility)**.** *A metabolic network $\mathcal{N}$ can operate at the Causal Firewall threshold $\sigma = \lambda\, \tau_d\, \ell^3 = 1$ only if its coarse-grained assembly depth satisfies:*

$$\tilde{A}_{\mathrm{RA}}(\mathcal{N}) \;\leq\; \frac{\tau_{\mathrm{cycle}}}{\tau_{\mathrm{irrev}}}, \tag{7}$$

where τ_{cycle} is the cell cycle time and τ_{irrev} is the mean time per irreversible reaction step.

Proof. By assumption, M^* is essential to division — the network cannot complete replication without producing M^*. By the critical path definition of RAHC [Sandeman, 2026e], producing M^* requires traversing at least $\tilde{A}_{\mathrm{RA}}(\mathcal{N})$ sequential irreversible steps, each taking time τ_{irrev}. Therefore $\tau_{\mathrm{cycle}} \geq \tilde{A}_{\mathrm{RA}}(\mathcal{N}) \cdot \tau_{\mathrm{irrev}}$, which rearranges to (7). For the Causal Firewall threshold $\sigma = 1$ to hold, the system requires exactly one new actualization event per coherence volume per decoherence time τ_d. If $\tilde{A}_{\mathrm{RA}} \cdot \tau_{\mathrm{irrev}} > \tau_{\mathrm{cycle}}$, the system cannot sustain the $\sigma = 1$ causal flux, and the network falls below the Causal Firewall. The bound (7) is therefore necessary for Firewall compatibility. □

This proposition gives the Causal Firewall a direct operational interpretation in terms of metabolic network structure. It says: a cell can live near the edge of chaos only if its most causal-depth-intensive pathway fits inside its cell cycle. Cells whose critical path is too long relative to their cycle time cannot maintain the causal coherence required for biological complexity.

Three consequences. **(i) The Firewall selects for short critical paths.** Organisms near the Causal Firewall threshold are simultaneously under selection pressure to minimise $\tilde{A}_{\mathrm{RA}}(\mathcal{N})$, since a shorter critical path both increases replication rate (Proposition 5.5(iii) of RAHC) and relaxes the constraint (7). The two selection pressures — for fast replication and for Firewall compatibility — are *the same pressure expressed at different scales*. At the molecular scale, it is selection on $A_{\mathrm{RA}}(M)$;

at the network scale, it is selection on $\tilde{A}_{\mathrm{RA}}(\mathscr{N})$; at the organism scale, it is selection on cell cycle time. One causal mechanism drives all three.

(ii) A quantitative origin-of-life constraint. The first self-replicating systems must have satisfied both $\sigma \geq 1$ (the Causal Firewall) and equation (7). Combined with the exponential suppression of replication rate by $A_{\mathrm{RA}}(M)$ (equation from §4.6), this gives a joint constraint on the space of viable first replicators — the sandwich bound of equation (1) (proved as Proposition 4.4):

$$A(M_0^*) \;\leq\; \tilde{A}_{\mathrm{RA}}(\mathscr{N}_0) \;\leq\; \frac{\tau_{\mathrm{cycle},0}}{\tau_{\mathrm{irrev}}},$$

where $\mathscr{N}_0$ is the proto-metabolic network and M_0^* is its most complex essential product. This double bound bounds the Assembly Index of the first replicators from both above and below — a quantitative, falsifiable constraint on the character of the origin of life.

(iii) Unification of the $\mu = 1$ transitions. The same $\mu = 1$ Erdős-Rényi transition governs QCD confinement, galactic rotation curve onset, the quantum fault-tolerance threshold, and — now via equation (7) — the Causal Firewall. In each case the transition is between a regime where causal connectivity is insufficient to maintain coherent structure ($\mu < 1$) and one where a giant causal component exists ($\mu > 1$). The biological Causal Firewall is the $\mu = 1$ transition applied to the molecular causal graph. The coarse-grained assembly depth is the quantity that determines how close a given metabolic network sits to this universal threshold.

4.6.5 Resolution via Erdős-Rényi: Hard Wall 1 is Proved

The Causal Firewall threshold $\sigma = \lambda\, \tau_d\, \ell^3 = 1$ is not a conjecture requiring an independent limit theorem. It is a direct consequence of the Poisson-CSG generative measure (RAGC Derivation 1, [Sandeman, 2026b]) combined with the classical Erdős-Rényi percolation theorem.

The identification. The Poisson-CSG mean-field parameter is $\mu = \lambda\, V_{\mathrm{coh}}\, p_{\mathrm{th}}$, where $V_{\mathrm{coh}} = \ell^3$ is the pre-actualization coherence volume and p_{th} is the per-link actualization probability. Identifying $p_{\mathrm{th}} = \tau_d/\tau_{\mathrm{mol}}$ (the probability that a given molecular interaction crosses the kinematic threshold within the decoherence time) gives $\mu = \lambda\, \tau_d\, \ell^3 = \sigma$ exactly. This identification is dimensionally consistent and physically natural: p_{th} in the quantum computing context is precisely the fraction of interactions that are on-shell within the syndrome extraction cycle, which in the biological context maps to $\tau_d/\tau_{\mathrm{mol}}$.

Erdős-Rényi applied to the causal graph. The RA causal graph at scale ℓ is a directed random graph in which each vertex connects to a mean of $\mu = \lambda\, V_{\mathrm{coh}}\, p_{\mathrm{th}}$ causal predecessors. The directed Erdős-Rényi theorem [Erdős & Rényi, 1960] states that a random directed graph $G(N, p = \mu/N)$ has:

- $\mu < 1$: all strongly connected components are $O(\ln N)$ — the graph fragments into trees, and no universal causal connectivity exists.
- $\mu = 1$: critical point; the largest strongly connected component scales as $N^{2/3}$.
- $\mu > 1$: a giant strongly connected component of size $\approx(1 - e^{-\mu})N$ emerges.

The shared causal ledger required by the Causal Firewall conditions is precisely a giant strongly connected component: a region in which every pair of vertices has a directed causal path in both directions, guaranteeing that all observers can reconstruct the same actualization history. Therefore:

- $\sigma < 1$ ($\mu < 1$): the causal graph fragments — no shared ledger — biological assembly of arbitrary depth is impossible.
- $\sigma > 1$ ($\mu > 1$): the giant causal component exists — shared ledger established — biological complexity is possible.

- $\sigma = 1$ ($\mu = 1$): *exactly* the Causal Firewall threshold.

The trace-distance convergence criterion (the bound on $\|\rho^{(i)}(t) - \rho^{(j)}(t)\|_{\mathrm{tr}}$) is now provable: the trace-distance bound between observer-dependent actualization histories scales as $\varepsilon \sim N^{-1/3}$ at criticality and decays exponentially for $\mu > 1$, inherited directly from the Erdős-Rényi scaling of the strongly connected component.

Remaining gap. The identification $p_{\mathrm{th}} = \tau_d/\tau_{\mathrm{mol}}$ requires a derivation from the RA-CSG dynamics showing that the per-link actualization probability in the biological molecular regime equals this decoherence ratio. This is a single, precisely-stated calculation in open quantum systems theory rather than the full limit theorem previously required. **Collaborator target:** open quantum systems, decoherence theory.

4.7 Derivation 3: The Markov Blanket and Physical Synchronization

Goal. Apply the Markov Blanket Isomorphism—proved in RAHC [Sandeman, 2026e], Theorems 2–3—to the biological context, establishing how organisms track the external causal ledger through physical synchronization.

4.7.1 The Markov Blanket—Formal Definition

In a Bayesian network $N = (X, E_N)$, the Markov blanket $\mathrm{MB}(S)$ of a node set S is the minimal set rendering S conditionally independent of all other nodes:

$$X_S \perp X_{V\setminus(S\cup \mathrm{MB}(S))} \mid X_{\mathrm{MB}(S)}. \tag{8}$$

For a biological organism S, $\mathrm{MB}(S)$ consists of sensory states (environment $\to S$ inputs), active states ($S \to$ environment outputs), and internal states not directly connected to the environment.

4.7.2 DAG Markov Property

In the RA causal DAG $G = (V, E)$, for a vertex v, define the past light cone $J^-(v) := \{u \in V : u \text{ causally precedes } v\} \cup \{v\}$, and its boundary $\partial J^-(v) := \{u \in J^-(v) : \exists w \notin J^-(v) \text{ with } (u,w) \in E\}$.

Theorem 4.5 (DAG Markov Property [Sandeman, 2026e], Theorem 2)**.** *In the causal DAG G with the RA-CSG probability measure, for any vertex v:*

$$\rho_v \perp \rho_{V\setminus J^-(v)} \,\Big|\, \rho_{\partial J^-(v)}. \tag{9}$$

4.7.3 The Isomorphism Theorem

Theorem 4.6 (Markov Blanket $\leftrightarrow$ Light Cone Boundary Isomorphism [Sandeman, 2026e], Theorem 3; Lean-verified core)**.** *Let O be an organism modeled as a subgraph $G_O \subseteq G$. There exists a coarse-graining map $\varphi : \partial J^-(G_O) \to \mathrm{MB}(G_O)$ such that:*

(a) *Sensory states correspond to incoming boundary vertices of $\partial J^-(G_O)$.*

(b) *Active states correspond to outgoing boundary vertices of G_O.*

(c) *The conditional independence structure is preserved:*

$$X_{G_O} \perp X_{V\setminus(G_O\cup\partial J^-(G_O))} \,\Big|\, X_{\partial J^-(G_O)}. \tag{10}$$

Proofs are given in RAHC [Sandeman, 2026e]. The biological interpretation follows: sensory states detect already-actualized classical facts encoded redundantly in environmental degrees of freedom; active states inject new actualization events into the external DAG.

4.7.4 Bayesian Updating as Physical Synchronization

Minimizing the variational free energy [Friston, 2010] $F[q] := \mathrm{KL}[q(x)\|p(x|o)] - \ln p(o)$ corresponds physically to executing the sequence of actualization events that synchronize the organism's internal DAG subgraph with the observed boundary state $\partial J^-(G_O)$. The full formal development of this structural correspondence—including the Generative Motif definition, the Tier 4 Transition Criterion, the Action Value function, and the explicit derivation from the Steering Cost Functional—is given in the companion paper RAHC [Sandeman, 2026e]. Derivation 4 below establishes the thermodynamic bounds that underpin that treatment.

4.8 Derivation 4: The Algorithmic Thermodynamics of Graph Steering [Lean-verified bound]

Goal. Derive the thermodynamic bounds on forced actualization (graph steering) and establish the Epistemic-Physical Equivalence that underpins the Action Value function in RAHC [Sandeman, 2026e].

4.8.1 Setup: Agency as Causal Forcing

Under passive RA-CSG dynamics, the adjacent possible $\mathrm{AP}_{\mathrm{free}}(G_n)$ contains all on-shell transitions available at step n. When an organism intervenes, it forces a restricted adjacent possible $\mathrm{AP}_{\mathrm{forced}}(G_n) \subsetneq \mathrm{AP}_{\mathrm{free}}(G_n)$. The degree of restriction is $\Delta\Omega := |\mathrm{AP}_{\mathrm{free}}| - |\mathrm{AP}_{\mathrm{forced}}| > 0$.

4.8.2 Thermodynamic Bounds on Causal Forcing

The following two theorems are proved in RAHC [Sandeman, 2026e], Theorems 4–5 (Lean-verified). We state them here as the thermodynamic foundation for the biological applications that follow.

Theorem 4.7 (Landauer Cost of Actualization [Sandeman, 2026e], Theorem 4; Lean-verified)**.** *Each irreversible actualization event that eliminates Δq bits of quantum uncertainty dissipates at minimum*

$$W_{\mathrm{act}} \;=\; \Delta q \cdot k_B T \cdot \ln 2. \tag{11}$$

Definition 4.8 (Steering Cost Functional)**.** For a forced actualization sequence $\pi = (v_1, \ldots, v_n)$:

$$W_{\mathrm{steer}}(\pi) \;:=\; \sum_{k=1}^{n} \Big[W_{\mathrm{act}}(v_k) + W_{\mathrm{sel}}(v_k) \Big], \tag{12}$$

where $W_{\mathrm{act}}(v_k) = \Delta q_k \cdot k_B T \cdot \ln 2$ and

$$W_{\mathrm{sel}}(v_k) \;=\; k_B T \cdot \ln\left(\frac{|\mathrm{AP}_{\mathrm{free}}(G_{k-1})|}{|\mathrm{AP}_{\mathrm{forced}}(G_{k-1})|} \right). \tag{13}$$

Theorem 4.9 (Epistemic-Physical Equivalence [Sandeman, 2026e], Theorem 5; Lean-verified)**.** *Under the RA-DAG model:*

$$W_{\mathrm{steer}}(\pi) \;\geq\; k_B T \cdot \Delta F[q], \tag{14}$$

where $\Delta F[q]$ is the reduction in variational free energy (in nats) achieved by π.

The organism cannot reduce its uncertainty about the world for free. Every bit of prediction accuracy comes at a thermodynamic cost.

Connection to dissipative adaptation. Dissipative adaptation [England, 2013] selects for structures that minimize W_{steer} for a given ΔF—acquiring predictive accuracy at minimum thermodynamic cost. In RA, this is simultaneously selection for: minimal causal assembly depth A_{RA} (Derivation 1), operation at the edge of chaos ($\sigma = 1$, Derivation 2), and faithful internal DAG synchronization with the external ledger (Derivation 3). All four derivations are facets of the same optimization. RAHC [Sandeman, 2026e] extends this result to the full four-tier complexity hierarchy, showing that the same thermodynamic bounds that govern biological assembly also govern cognition and agentic graph steering at every scale.

4.8.3 Hard Wall

Theorem 4.9 establishes a lower bound but not a tight one. Achieving the Landauer bound requires quasi-static (infinitely slow) state changes; real biological systems operate far from quasi-static conditions. The excess dissipation $W_{\text{steer}} - k_B T \cdot \Delta F$ is system-specific. Computing Ω_{free} also requires knowing the RA-CSG transition probabilities (the measure problem from RAGC Derivation 1, inherited here).

4.9 Summary: Status of Each Derivation

Derivation 1 (Assembly Index). *Established with computational and theoretical support.* $A_{\text{RA}}(M) \geq A(M)$ (Theorem 4.2, proved in RAHC [Sandeman, 2026e]); load-bearing Condition C1′ established via variational argument and confirmed by B3LYP/6-311+G* DFT survey across all five primary bond types (Table 1); non-ergodicity formalized. Remaining scope: polymer assembly regime with many-body effects.

Derivation 2 (Causal Firewall). *Conjecture requiring a limit theorem.* Threshold equation $\lambda \tau_d \ell^3 = 1$ stated; England compatibility derived; evolutionary predictions P2–P3 formulated. Hard wall: limit theorem with explicit error bound in trace distance or relative entropy.

Derivation 3 (Markov Blanket). *Theorems proved in RAHC [Sandeman, 2026e]; **Lean-verified core**.* DAG Markov Property (Theorem 4.5) and Markov Blanket Isomorphism (Theorem 4.6) cited from RAHC; biological interpretation and synchronization protocol developed here. Hard wall: physical substrate of the generative model.

Derivation 4 (Steering Cost). *Theorems proved in RAHC [Sandeman, 2026e]; **Lean-verified**.* Landauer Cost (Theorem 4.7) and Epistemic-Physical Equivalence (Theorem 4.9) cited from RAHC; biological applications (England bound, evolutionary predictions) developed here. All four derivations are facets of the same optimization: minimal causal assembly depth, edge-of-chaos operation, faithful DAG synchronization, and minimum steering cost.

5 Conclusion: A Unified Account of Complexity and Life

By adopting the foundational ontology of Relational Actualism, the conceptual discontinuity between quantum mechanics, thermodynamics, and biology is resolved without appeal to new constants, special initial conditions, or anthropic reasoning. The framework accomplishes this through a single commitment: the irreversible actualization of quantum states is the primitive physical event from which all macroscopic structure is built.

The four mathematical derivations above establish a unified framework in which the Actualism Assembly Index provides the causal measure of biological complexity; the Causal Firewall defines the physical regime where that complexity can be stably preserved; the Markov blanket isomorphism explains how organisms track the external causal ledger; and the Steering Cost Functional and Epistemic-Physical Equivalence bound quantify the thermodynamic price of causal forcing. All four are facets of a single optimization: natural selection is selection for organisms that acquire predictive accuracy about the causal ledger at minimum thermodynamic cost, via the shallowest causal assembly pathways, at the edge of chaos.

The extension of these thermodynamic bounds to the full hierarchy of physical complexity—from fundamental interactions through dissipative structures, autopoietic systems, and agentic intelligence, including the Generative Motif criterion and the correspondence with Friston's Free Energy Principle—is developed in the companion paper RAHC [Sandeman, 2026e]. RACI establishes the physics; RAHC deploys it.

An unexpected structural parallel: the Causal Firewall threshold in RACI (the phase transition between abiotic chemistry and stable biological complexity) is the biological counterpart of the critical actualization density λ_c in the companion paper RADM [Sandeman, 2026f]. In RADM,

λ_c marks where the causal DAG transitions from 3D-embeddable (dense, structured) to tree-like (sparse, dimensionally reduced). Both thresholds are phase transitions in the same causal graph, at different scales of coarse-graining.

The five empirical predictions of Section 4 follow from this framework and are testable now, independently of whether the remaining mathematical derivations are complete:

1. The distribution of life in the universe is physically constrained by the Causal Firewall conditions, yielding a habitability criterion more fundamental than liquid water or temperature.
2. Natural selection for replication speed is simultaneously selection for minimal causal assembly depth, measurable across metabolic networks using Assembly Theory tools.
3. The first replicators were the causally shallowest self-replicating systems possible — a constraint evaluable with existing Assembly Theory tools via the sandwich bound.
4. Boltzmann Brains are categorically, not merely statistically, impossible: the RA account replaces a probabilistic suppression with a causal impossibility.
5. The search for extraterrestrial life should be weighted toward environments satisfying the substrate-independent Causal Firewall criteria, not merely toward chemically favourable circumstellar habitable zones.

The universe builds its history one irreversible actualization at a time. Life is not an anomaly within that history; it is its most elaborate physical expression at the biological scale — organisms that maintain stable causal closure against thermodynamic dissolution, assemble complexity through irreversible causal depth, and operate at the precise edge where actualization rate and environmental disruption balance. What those organisms then do with that stability — how they model, predict, and steer the causal ledger that made them — is the subject of the companion paper RAHC [Sandeman, 2026e].

The complete RA suite also includes the Engine of Becoming paper (RAEB [Sandeman, 2026RAEB]), the particle classification and Standard Model topology paper (RATM [Sandeman, 2026RATM]), the field equation uniqueness paper (RACL [Sandeman, 2026RACL]), and the programme overview (Foundation [Sandeman, 2026g]).

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